Kinora Ankle

Clinical AI Orientation (Part 10)

Structured reference for RAG / AI-assisted physiotherapy consultation

Bones, Joints, Ligaments, Tendons, Muscles, Biomechanics, Stability, Neurovascular, Assessment, Pathologies & Rehabilitation

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1. Bones

RECORD: Talus

ID: ANK-BONE-01

Bone: Talus

Landmarks: Head, neck, body; trochlea; lateral/medial/posterior processes; sulcus tali; sinus tarsi contribution

Articulations: Tibia/fibula (talocrural); calcaneus (subtalar); navicular (talonavicular)

Muscle Attachments: No direct muscle attachments — unique among tarsals; ligamentous only

Ligament Attachments: ATFL, PTFL, CFL (via calcaneus link), deltoid components, interosseous talocalcaneal, spring ligament supports head

Blood Supply: Posterior tibial, anterior tibial, fibular anastomoses — vulnerable watershed at neck (AVN risk)

Innervation: Deep peroneal, tibial articular branches

Ossification: Primary center in utero; ossifies early; accessory os trigonum common posterior

Biomechanics: Keystone of ankle mortise and STJ; transmits tibial load to calcaneus/forefoot; 3D motion hinge

Clinical Importance: Osteochondral lesions (OLT); talar neck fracture AVN risk (Hawkins); os trigonum posterior impingement; lateral process snowboarder's fracture

Imaging: XR AP/mortise/lateral; CT fracture mapping; MRI cartilage/AVN/OCD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Calcaneus

ID: ANK-BONE-02

Bone: Calcaneus

Landmarks: Tuberosity, sustentaculum tali, anterior process, facets (ant/mid/post), calcaneal sulcus, peroneal trochlea

Articulations: Talus (STJ); cuboid (calcaneocuboid)

Muscle Attachments: Achilles to tuberosity; abductor hallucis/FDB/ADM origins; quadratus plantae; extensor digitorum brevis (dorsal)

Ligament Attachments: CFL, spring (sustentaculum), long/short plantar, interosseous talocalcaneal, lateral/medial talocalcaneal

Blood Supply: Posterior tibial, fibular, medial/lateral plantar branches; rich periosteal supply

Innervation: Medial/lateral plantar, sural, tibial articular

Ossification: Ossifies mid-fetal life; apophysis (Sever disease site) fuses adolescence

Biomechanics: Accepts majority of talar load; lever for Achilles; STJ facets guide inversion/eversion

Clinical Importance: Calcaneal fracture (Bohler/Gissane angles); plantar fasciopathy origin; Sever apophysitis; Haglund deformity

Imaging: XR lateral/axial Harris; CT trauma; MRI fasciopathy/Haglund

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Distal Tibia

ID: ANK-BONE-03

Bone: Distal Tibia

Landmarks: Medial malleolus, plafond (pilon), Chaput tubercle (AITFL), Tillaux region (adolescents), posterior malleolus (Volkmann)

Articulations: Talus (plafond/medial malleolus); fibula (syndesmosis)

Muscle Attachments: None major at plafond; TP/FDL/FHL tendons posterior; TA anterior

Ligament Attachments: Deltoid to medial malleolus; AITFL/PITFL/interosseous syndesmosis; ankle capsule

Blood Supply: Anterior/posterior tibial, malleolar anastomoses

Innervation: Saphenous (medial), deep peroneal articular, tibial

Ossification: Distal tibial physis — Tillaux/triplane fracture patterns in adolescents

Biomechanics: Forms medial/superior mortise; plafond cartilage critical; length/rotation affect talar congruence

Clinical Importance: Pilon fracture; medial malleolus fracture; syndesmotic avulsions; ankle OA after malunion

Imaging: XR AP/mortise/lateral; CT articular; MRI ligaments

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Distal Fibula

ID: ANK-BONE-04

Bone: Distal Fibula

Landmarks: Lateral malleolus, malleolar fossa, Wagstaffe tubercle (AITFL)

Articulations: Talus laterally; tibia via syndesmosis

Muscle Attachments: Peroneus longus/brevis posterior; FHL nearby

Ligament Attachments: ATFL, CFL, PTFL; AITFL/PITFL; superior peroneal retinaculum

Blood Supply: Fibular artery malleolar branches; peroneal perforators

Innervation: Superficial/deep peroneal, sural articular

Ossification: Distal fibular physis — common pediatric ankle fracture site

Biomechanics: Lateral mortise wall; length restores lateral talar constraint (Weber/Lauge-Hansen)

Clinical Importance: Weber A/B/C fractures; Maisonneuve association; ligament avulsions

Imaging: Full-length tib-fib if Maisonneuve suspected; mortise view clear space

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Navicular

ID: ANK-BONE-05

Bone: Navicular

Landmarks: Tuberosity (TP insertion), articular facets for talus and cuneiforms

Articulations: Talus, three cuneiforms, occasionally cuboid

Muscle Attachments: Tibialis posterior primary; spring ligament complex support

Ligament Attachments: Spring (plantar calcaneonavicular), dorsal talonavicular, cuneonavicular ligaments

Blood Supply: Dorsalis pedis branches; medial plantar

Innervation: Deep peroneal, medial plantar

Ossification: Ossifies ~age 3; accessory navicular (os tibiale externum) common medial

Biomechanics: Keystone medial longitudinal arch; TP inserts — PTTD critical

Clinical Importance: Navicular stress fracture (high-risk); accessory navicular syndrome; Kohler disease (pediatric osteochondrosis)

Imaging: XR AP/oblique/lateral; MRI stress; CT nonunion

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Cuboid

ID: ANK-BONE-06

Bone: Cuboid

Landmarks: Peroneal groove (PL tendon); calcaneal and metatarsal facets

Articulations: Calcaneus, lateral cuneiform, MT4-5, occasionally navicular

Muscle Attachments: Peroneus longus groove; FHB origins nearby; EDB

Ligament Attachments: Long/short plantar, calcaneocuboid, cuboideonavicular, TMT ligaments

Blood Supply: Lateral tarsal / arcuate artery branches

Innervation: Lateral plantar, sural, deep peroneal

Ossification: Ossifies ~age 9 prenatal/early — primary center early childhood

Biomechanics: Lateral column stability; PL mechanical advantage under cuboid

Clinical Importance: Cuboid syndrome (subluxation concept); fracture; PL tendinopathy at groove

Imaging: XR oblique; MRI/US PL; clinical cuboid whip assessment

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Medial Cuneiform

ID: ANK-BONE-07

Bone: Medial Cuneiform

Landmarks: Largest cuneiform; PL and TA insertions; 1st TMT base

Articulations: Navicular, intermediate cuneiform, 1st MT, occasionally 2nd MT

Muscle Attachments: Tibialis anterior, peroneus longus, partially FHB

Ligament Attachments: Lisfranc complex medial; intercuneiform; cuneonavicular

Blood Supply: Dorsal/plantar arterial network

Innervation: Deep peroneal, medial plantar

Ossification: Ossifies ~age 2

Biomechanics: Medial column; 1st ray stability; TA/PL force couple at medial cuneiform

Clinical Importance: Lisfranc injury medial; 1st TMT OA; bunion surgery osteotomies involve region

Imaging: Weight-bearing XR; CT Lisfranc

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Intermediate Cuneiform

ID: ANK-BONE-08

Bone: Intermediate Cuneiform

Landmarks: Smallest cuneiform; recessed — keystone of transverse arch

Articulations: Navicular, medial/lateral cuneiforms, 2nd MT

Muscle Attachments: Minimal direct; interossei nearby

Ligament Attachments: Lisfranc ligament attaches from medial cuneiform to 2nd MT base — critical

Blood Supply: Dorsal arterial network

Innervation: Deep peroneal, plantar nerves

Ossification: Ossifies ~age 3

Biomechanics: Recessed position locks 2nd MT — Lisfranc stability cornerstone

Clinical Importance: Lisfranc ligament injury — subtle diastasis 1st-2nd MT

Imaging: WB XR with comparison; CT/MRI Lisfranc

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lateral Cuneiform

ID: ANK-BONE-09

Bone: Lateral Cuneiform

Landmarks: Articulates cuboid and 3rd MT

Articulations: Navicular, intermediate cuneiform, cuboid, 3rd MT

Muscle Attachments: FHB / adductor related plantar; interossei

Ligament Attachments: Intercuneiform, TMT, cuboideocuneiform ligaments

Blood Supply: Arcuate / lateral tarsal branches

Innervation: Deep peroneal, lateral plantar

Ossification: Ossifies ~age 1

Biomechanics: Lateral midfoot column; TMT stability

Clinical Importance: Lisfranc lateral column; midfoot sprain/OA

Imaging: WB XR midfoot series; CT

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Joints

RECORD: Talocrural Joint

ID: ANK-JNT-01

Joint: Talocrural Joint

Type: Synovial hinge (ginglymus) — primarily sagittal DF/PF with coupled minor rotation

Capsule: Capsule thin anterior/posterior; reinforced by collaterals medially/laterally

Articular Surfaces: Tibial plafond + malleoli with talar trochlea (wider anteriorly)

Ligaments: ATFL, CFL, PTFL laterally; deltoid medially; syndesmosis proximally related

Blood Supply: Malleolar arterial anastomosis

Innervation: Deep peroneal, tibial, saphenous, sural articular branches

Arthrokinematics: Convex talus rolls/glides on concave tibia — DF: posterior glide of talus; PF: anterior glide (open chain)

Osteokinematics: DF ~20 deg; PF ~50 deg typical (individual); slight ABD/ADD and rotation coupled

Degrees of Freedom: Primarily 1 DOF (DF/PF) with accessory motions

Stability: Bony mortise congruence + ligaments; most stable in DF (wider anterior talus)

Biomechanics: Primary sagittal ankle rocker in gait; high compressive loads in run/jump

Clinical Tests: Anterior drawer, talar tilt, ROM DF/PF knee flexed/extended (Silfverskiöld for gastroc)

Pain Referral: Anterior/lateral/medial ankle; may refer to midfoot

Common Pathologies: Lateral sprain, OA, impingement, OLT, fracture-dislocation

Rehabilitation: POLICE/PEACE&LOVE early; restore DF; progressive strength/balance; Ottawa rules for imaging

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Subtalar Joint

ID: ANK-JNT-02

Joint: Subtalar Joint

Type: Synovial plane/compound (anterior and posterior articulations) — triplanar motion

Capsule: Capsule + interosseous talocalcaneal ligament in sinus tarsi

Articular Surfaces: Talar facets on calcaneal posterior/middle/anterior facets

Ligaments: Interosseous talocalcaneal, cervical, CFL (stabilizes STJ laterally), deltoid deep fibers, spring supports talar head

Blood Supply: Posterior tibial / fibular branches to sinus tarsi

Innervation: Tibial, deep peroneal articular

Arthrokinematics: Screw-like motion of talus on calcaneus; open-chain calcaneal INV/EV; closed-chain tibial rotation coupling

Osteokinematics: Inversion ~20 deg / eversion ~10 deg approximate total STJ contribution

Degrees of Freedom: 1 primary functional DOF along oblique axis (triplanar pronation/supination)

Stability: Interosseous ligament + CFL + bony facets; sinus tarsi stability

Biomechanics: Converts tibial rotation to foot pronation/supination; shock absorption early stance

Clinical Tests: Talar tilt (also TC), STJ medial/lateral glide, Coleman block for cavovarus workup

Pain Referral: Sinus tarsi / lateral hindfoot pain common

Common Pathologies: Sinus tarsi syndrome, STJ OA, coalition, calcaneal fracture malunion

Rehabilitation: Restore INV/EV control; TP/fibularis balance; proprioception

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Distal Tibiofibular Syndesmosis

ID: ANK-JNT-03

Joint: Distal Tibiofibular Syndesmosis

Type: Fibrous syndesmosis (amphiarthrosis) with synovial recess superiorly variable

Capsule: Not a classic capsule — ligaments + interosseous membrane

Articular Surfaces: Convex fibula against concave tibial fibular notch

Ligaments: AITFL, PITFL, interosseous ligament/membrane, inferior transverse ligament

Blood Supply: Peroneal / anterior tibial perforators

Innervation: Deep peroneal, tibial articular

Arthrokinematics: Slight fibular translation/ER with DF to accommodate wide anterior talus

Osteokinematics: Minimal osteokinematic motion — essential for mortise flexibility

Degrees of Freedom: Accessory micromotion (not primary DOF)

Stability: Ligamentous complex critical — disruption widens mortise and destabilizes talus

Biomechanics: Maintains mortise integrity under ER/DF load; high ankle sprain mechanism

Clinical Tests: Squeeze, ER stress, Cotton, crossed-leg; syndesmosis tenderness

Pain Referral: Anterolateral ankle above joint line

Common Pathologies: High ankle sprain; Maisonneuve; chronic instability/OA if missed

Rehabilitation: Longer protection than lateral sprain; progressive DF/ER loading; surgery if unstable diastasis

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Talonavicular Joint

ID: ANK-JNT-04

Joint: Talonavicular Joint

Type: Synovial ball-and-socket–like (part of acetabulum pedis / TN-CC complex)

Capsule: Capsule continuous with midfoot

Articular Surfaces: Talar head with navicular concavity

Ligaments: Spring ligament (plantar support), dorsal talonavicular, bifurcate, deltoid tibionavicular

Blood Supply: Dorsalis pedis / medial plantar branches

Innervation: Deep peroneal, medial plantar

Arthrokinematics: Large mobility — key to midfoot adaptability; TN motion major in foot pronation/supination

Osteokinematics: Triplanar midfoot motion contribution

Degrees of Freedom: Multiplanar — functionally linked to STJ (acetabulum pedis)

Stability: Spring ligament + TP tendon dynamic support of talar head

Biomechanics: Medial column motion; failure → adult-acquired flatfoot with PTTD/spring attenuation

Clinical Tests: Single heel rise (TP), too-many-toes, navicular drop

Pain Referral: Medial midfoot / arch

Common Pathologies: PTTD, spring ligament tear, TN OA, Kohler/accessory navicular

Rehabilitation: Arch support orthoses selected; TP progressive loading; surgical reconstruction advanced PTTD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Calcaneocuboid Joint

ID: ANK-JNT-05

Joint: Calcaneocuboid Joint

Type: Synovial saddle

Capsule: Capsule reinforced by plantar/dorsal ligaments

Articular Surfaces: Anterior calcaneus with posterior cuboid

Ligaments: Long and short plantar ligaments, bifurcate (calcaneocuboid part), dorsal CC ligaments

Blood Supply: Lateral tarsal branches

Innervation: Lateral plantar, sural, deep peroneal

Arthrokinematics: Limited gliding; lateral column stability; locks with midtarsal during late stance

Osteokinematics: Minimal INV/EV contribution vs TN

Degrees of Freedom: Limited DOF vs TN

Stability: Plantar ligaments + PL tendon dynamic

Biomechanics: Lateral column length/stability; Chopart joint with TN forms transverse tarsal

Clinical Tests: Cuboid mobility tests; PL assessment

Pain Referral: Lateral midfoot

Common Pathologies: Cuboid syndrome, CC OA, PL pathology

Rehabilitation: Lateral column mobilization/stability; PL loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Transverse Tarsal (Chopart) Joint

ID: ANK-JNT-06

Joint: Transverse Tarsal (Chopart) Joint

Type: Functional compound joint — talonavicular + calcaneocuboid

Capsule: Combined capsules/ligaments of TN and CC

Articular Surfaces: As TN + CC

Ligaments: Spring, bifurcate, long/short plantar, dorsal midtarsal ligaments

Blood Supply: Dorsalis pedis / lateral tarsal

Innervation: Deep peroneal, medial/lateral plantar

Arthrokinematics: Axes allow flexibility early stance and rigidity late stance (oblique + longitudinal midtarsal axes model — Elftman/Manter classical teaching)

Osteokinematics: Pronation unlocks / supination locks midfoot for push-off (simplified clinical model)

Degrees of Freedom: Coupled multiplanar

Stability: Ligamentous + TP/PL dynamic

Biomechanics: Critical transition flexibility→rigidity in gait

Clinical Tests: Midtarsal mobility; arch assessment

Pain Referral: Midfoot

Common Pathologies: Chopart injury, midfoot sprain, flatfoot/cavus compensation

Rehabilitation: Restore control of pronation/supination timing; orthoses; progressive loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Ligaments

RECORD: ATFL (Anterior Talofibular Ligament)

ID: ANK-LIG-01

Ligament: ATFL (Anterior Talofibular Ligament)

Origin: Anterior distal fibula (lateral malleolus)

Insertion: Neck of talus anterolaterally

Fiber Orientation: Horizontal / slightly inferior when ankle neutral — most horizontal of lateral ligaments

Function: Primary restraint to anterior talar translation and inversion in PF

Biomechanics: First and most commonly injured lateral ligament; taut in PF+INV

Injury Mechanism: Plantarflexion-inversion sprain (classic lateral ankle sprain)

Healing: Extra-articular — usually good healing with protected early motion for grade I-II

Clinical Tests: Anterior drawer (ankle PF ~10-20 deg); anterolateral palpation

Imaging: MRI/US for grade; XR Ottawa rules for fracture

Rehabilitation: Early protected ROM, progressive strength, balance/NMC; brace for return

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: CFL (Calcaneofibular Ligament)

ID: ANK-LIG-02

Ligament: CFL (Calcaneofibular Ligament)

Origin: Tip of lateral malleolus

Insertion: Lateral calcaneus

Fiber Orientation: Cord-like, inferior-posterior — crosses both TC and STJ

Function: Resists inversion in neutral/DF; stabilizes STJ as well as TC

Biomechanics: Taut in DF+INV; injured with ATFL in higher-grade sprains

Injury Mechanism: Inversion with more DF component; grade II-III lateral sprain

Healing: Extra-articular — heals with bracing/early motion typically

Clinical Tests: Talar tilt (inversion stress); STJ tilt contribution

Imaging: MRI/US; stress XR historical

Rehabilitation: Frontal-plane strength (fibularis); STJ control; progressive COD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: PTFL (Posterior Talofibular Ligament)

ID: ANK-LIG-03

Ligament: PTFL (Posterior Talofibular Ligament)

Origin: Malleolar fossa of fibula

Insertion: Lateral tubercle posterior talar process

Fiber Orientation: Horizontal deep posterior

Function: Strongest lateral ligament; resists posterior talar translation / DF extremes

Biomechanics: Rarely isolated injury — usually with severe/dislocation patterns

Injury Mechanism: High-energy inversion/DF or dislocation

Healing: Severe injuries — often surgical ankle context

Clinical Tests: Posterior drawer rarely isolated clinically

Imaging: MRI

Rehabilitation: Per severe ankle trauma protocols

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deltoid Ligament Complex

ID: ANK-LIG-04

Ligament: Deltoid Ligament Complex

Origin: Medial malleolus (anterior/posterior colliculi)

Insertion: Navicular (tibionavicular), spring/sustentaculum (tibiocalcaneal), talus (deep anterior/posterior tibiotalar) — superficial and deep layers

Fiber Orientation: Fan-shaped medial — superficial spans to navicular/calcaneus; deep to talus

Function: Primary restraint to eversion and lateral talar translation; deep layer strongest against abduction of talus

Biomechanics: Stabilizes medial mortise; injured in eversion/ER; associated with Weber C / Maisonneuve

Injury Mechanism: Eversion or ER force; often with fibular fracture/syndesmosis

Healing: Deep deltoid injury may need longer protection; instability with ER stress

Clinical Tests: Eversion stress; Kleiger (ER) test; medial clear space on mortise XR

Imaging: MRI; mortise XR medial clear space >4-5 mm suggests deltoid/syndesmosis injury (context-dependent)

Rehabilitation: Protect eversion; progressive medial strength (TP); address syndesmosis if combined

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Anterior Inferior Tibiofibular Ligament (AITFL)

ID: ANK-LIG-05

Ligament: Anterior Inferior Tibiofibular Ligament (AITFL)

Origin: Chaput tubercle (tibia)

Insertion: Wagstaffe tubercle (fibula)

Fiber Orientation: Oblique inferior-lateral

Function: Primary anterior syndesmotic stabilizer

Biomechanics: Resists ER and posterior fibular translation; injured in high ankle sprain

Injury Mechanism: ER of foot with DF; contact; ski/boot; cutting with planted foot

Healing: Longer healing than ATFL — often 6-8+ weeks before cutting

Clinical Tests: Squeeze, ER stress, tenderness over AITFL

Imaging: MRI; WB XR tibiofibular clear space; stress views

Rehabilitation: Protect ER/DF early; progressive loading; surgery if diastasis unstable

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Posterior Inferior Tibiofibular Ligament (PITFL)

ID: ANK-LIG-06

Ligament: Posterior Inferior Tibiofibular Ligament (PITFL)

Origin: Posterior distal tibia (Volkmann region)

Insertion: Posterior distal fibula

Fiber Orientation: Strong posterior band + inferior transverse ligament contribution

Function: Strongest syndesmotic ligament; resists posterior diastasis

Biomechanics: Often intact as hinge in some injury patterns; avulsion with posterior malleolus fracture

Injury Mechanism: High-energy ER/DF; fracture-associated

Healing: Healing tied to fracture/syndesmosis fixation protocols

Clinical Tests: ER stress; posterior palpation

Imaging: MRI/CT for posterior malleolus

Rehabilitation: Per syndesmosis/ORIF protocols

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Interosseous Ligament / Membrane (Distal)

ID: ANK-LIG-07

Ligament: Interosseous Ligament / Membrane (Distal)

Origin: Length of tibia-fibula via IO membrane; distal IO ligament thickening

Insertion: Corresponding fibular border

Fiber Orientation: Vertical fibrous sheet

Function: Major syndesmotic stabilizer; load transfer tibia-fibula

Biomechanics: Disruption allows mortise widening; Maisonneuve tears proximal membrane

Injury Mechanism: ER force propagating up IO membrane

Healing: Requires anatomic reduction if unstable

Clinical Tests: Clinical syndesmosis cluster; full-length XR

Imaging: MRI; proximal fibula XR

Rehabilitation: Longer protection; avoid early aggressive ER/DF loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Spring Ligament (Plantar Calcaneonavicular)

ID: ANK-LIG-08

Ligament: Spring Ligament (Plantar Calcaneonavicular)

Origin: Sustentaculum tali of calcaneus

Insertion: Plantar navicular

Fiber Orientation: Superomedial, medioplantar, inferoplantar bands (complex)

Function: Supports talar head; static medial arch keystone support

Biomechanics: Works with TP tendon; failure with advanced PTTD → peritalar subluxation

Injury Mechanism: Degenerative attenuation with TP dysfunction; acute trauma less common isolated

Healing: Poor spontaneous restoration if elongated — may need reconstruction with TP surgery

Clinical Tests: Single heel rise, too-many-toes, arch assessment

Imaging: MRI spring ligament edema/discontinuity

Rehabilitation: Orthoses, TP loading early stages; surgical reconstruction selected advanced flatfoot

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Tendons

RECORD: Achilles Tendon

ID: ANK-TEN-01

Tendon: Achilles Tendon

Muscle: Gastrocnemius + soleus (triceps surae); plantaris variable

Origin: Musculotendinous junction mid-calf

Insertion: Calcaneal tuberosity

Innervation: Tibial nerve (S1-S2)

Blood Supply: Posterior tibial / peroneal — watershed 2-6 cm above insertion

Function: Primary plantarflexion torque transmitter

Biomechanics: Energy storage-release in running; highest load tendon in body (multiples of BW)

Clinical Importance: Tendinopathy continuum midportion vs insertional; rupture classic middle-aged athlete

Common Injuries: Midportion tendinopathy; insertional; complete rupture

Assessment: Palpation, Royal London, painful arc, Thompson for rupture, Silfverskiöld

Rehabilitation: Progressive loading (Alfredson/heavy slow/isometrics); rupture: functional rehab vs surgery shared decision

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Posterior Tendon

ID: ANK-TEN-02

Tendon: Tibialis Posterior Tendon

Muscle: Tibialis posterior

Origin: IO membrane / posterior tibia-fibula

Insertion: Navicular tuberosity + broad plantar inserts

Innervation: Tibial nerve (L4-L5)

Blood Supply: Posterior tibial artery

Function: Inversion, PF assist, dynamic medial arch support

Biomechanics: Eccentric control of pronation midstance; key in PTTD continuum

Clinical Importance: Most important dynamic arch supporter; dysfunction → adult-acquired flatfoot

Common Injuries: Tendinopathy, tenosynovitis, rupture (degenerative)

Assessment: Single heel rise (lack of inversion), too-many-toes, swelling behind medial malleolus

Rehabilitation: Stage-based: orthoses, TP progressive loading early; surgery advanced stages

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Anterior Tendon

ID: ANK-TEN-03

Tendon: Tibialis Anterior Tendon

Muscle: Tibialis anterior

Origin: Lateral tibia / IO membrane

Insertion: Medial cuneiform + 1st MT base

Innervation: Deep peroneal (L4-L5)

Blood Supply: Anterior tibial artery

Function: DF and inversion

Biomechanics: Eccentric heel-strike control; swing clearance

Clinical Importance: Foot drop if nerve injured; rare spontaneous rupture in older adults

Common Injuries: Tendinopathy; rupture; laceration

Assessment: Heel walk, DF strength, palpation anterior ankle

Rehabilitation: Eccentric DF loading; gait; AFO if neurogenic

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Longus Tendon

ID: ANK-TEN-04

Tendon: Peroneus Longus Tendon

Muscle: Peroneus (fibularis) longus

Origin: Proximal fibula

Insertion: Medial cuneiform + 1st MT base plantar (via cuboid groove)

Innervation: Superficial peroneal (L5-S1)

Blood Supply: Fibular artery perforators

Function: Eversion; PF 1st ray

Biomechanics: Turns at malleolus and cuboid — friction zones; balances TA at 1st ray

Clinical Importance: Tendinopathy; subluxation with retinaculum injury; cuboid interaction

Common Injuries: Tendinopathy, tears, dislocation

Assessment: Resisted eversion, retromalleolar palpation, DF-eversion subluxation test

Rehabilitation: Progressive eversion loading; address instability; surgery for recurrent dislocation

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Brevis Tendon

ID: ANK-TEN-05

Tendon: Peroneus Brevis Tendon

Muscle: Peroneus (fibularis) brevis

Origin: Distal fibula

Insertion: 5th MT tuberosity

Innervation: Superficial peroneal (L5-S1)

Blood Supply: Fibular perforators

Function: Eversion; lateral ankle dynamic stability

Biomechanics: Most commonly torn peroneal tendon (longitudinal splits)

Clinical Importance: Associated with lateral ankle instability and cavovarus

Common Injuries: Longitudinal tear, tendinopathy, 5th MT avulsion differential

Assessment: Eversion strength, US dynamic

Rehabilitation: Load + stability; repair selected tears

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Hallucis Longus Tendon

ID: ANK-TEN-06

Tendon: Flexor Hallucis Longus Tendon

Muscle: FHL

Origin: Posterior fibula

Insertion: Distal phalanx hallux

Innervation: Tibial nerve (S2-S3)

Blood Supply: Fibular/posterior tibial

Function: Hallux flexion; PF assist; arch assist

Biomechanics: Passes under sustentaculum; posteromedial ankle — dancer's tendinopathy / os trigonum overlap

Clinical Importance: Posterior impingement continuum in pointe/dance

Common Injuries: Tenosynovitis, trigger hallux, posteromedial pain

Assessment: Hallux flexion strength, posterior medial palpation

Rehabilitation: Load management; progressive hallux flexion; address os trigonum if indicated

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Digitorum Longus Tendon

ID: ANK-TEN-07

Tendon: Flexor Digitorum Longus Tendon

Muscle: FDL

Origin: Posterior tibia

Insertion: Distal phalanges 2-5

Innervation: Tibial nerve (S2-S3)

Blood Supply: Posterior tibial

Function: Toe flexion; PF/inversion assist

Biomechanics: Henry knot cross with FHL; tarsal tunnel contents

Clinical Importance: Tarsal tunnel differential; lesser toe deformities contribution

Common Injuries: Tenosynovitis uncommon isolated

Assessment: Toe flexion strength

Rehabilitation: Intrinsic/extrinsic balance training

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Extensor Hallucis Longus Tendon

ID: ANK-TEN-08

Tendon: Extensor Hallucis Longus Tendon

Muscle: EHL

Origin: Fibula / IO membrane

Insertion: Distal phalanx hallux dorsal

Innervation: Deep peroneal (L5)

Blood Supply: Anterior tibial

Function: Hallux extension; DF assist

Biomechanics: L5 myotome clinical testing

Clinical Importance: Laceration risk dorsum foot; L5 radiculopathy differential

Common Injuries: Laceration; rare rupture

Assessment: Hallux extension MMT

Rehabilitation: Protect/repair lacerations; progressive extension

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Extensor Digitorum Longus Tendon

ID: ANK-TEN-09

Tendon: Extensor Digitorum Longus Tendon

Muscle: EDL

Origin: Tibia/fibula/IO

Insertion: Extensor expansions toes 2-5

Innervation: Deep peroneal (L4-L5)

Blood Supply: Anterior tibial

Function: Toe extension; DF assist

Biomechanics: Swing clearance with TA/EHL

Clinical Importance: Anterior ankle structures under retinaculum

Common Injuries: Laceration; tenosynovitis rare

Assessment: Toe extension strength

Rehabilitation: DF endurance; retinacular protection post-op

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Muscles

RECORD: Gastrocnemius

ID: ANK-MUS-01

Muscle: Gastrocnemius

Origin: Femoral condyles

Insertion: Achilles → calcaneus

Innervation: Tibial S1-S2

Blood Supply: Popliteal sural branches

Function: PF + knee flexion; propulsion

EMG: High sprint/jump push-off

Clinical Importance: Tennis leg; Achilles load

Exercises: Straight-knee heel raises

Progressions: Isometrics → HSR → plyometrics

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Soleus

ID: ANK-MUS-02

Muscle: Soleus

Origin: Soleal line / fibula

Insertion: Achilles → calcaneus

Innervation: Tibial S1-S2

Blood Supply: Posterior tibial/fibular

Function: Primary postural PF; midstance tibial control

EMG: High continuous stance EMG

Clinical Importance: Achilles; soleus strain

Exercises: Bent-knee heel raises

Progressions: Heavy slow soleus bias → run hills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Anterior

ID: ANK-MUS-03

Muscle: Tibialis Anterior

Origin: Lateral tibia/IO

Insertion: Medial cuneiform/1st MT

Innervation: Deep peroneal L4-L5

Blood Supply: Anterior tibial

Function: DF + inversion; heel-strike control

EMG: High eccentric IC / swing concentric

Clinical Importance: Foot drop; MTSS ddx

Exercises: Heel walks, eccentric DF lowers

Progressions: Endurance → speed walking/run form

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Posterior

ID: ANK-MUS-04

Muscle: Tibialis Posterior

Origin: IO/posterior tib-fib

Insertion: Navicular + plantar tarsals

Innervation: Tibial L4-L5

Blood Supply: Posterior tibial

Function: Inversion + arch support

EMG: High midstance eccentric

Clinical Importance: PTTD key muscle

Exercises: Inversion band, short-foot, heel rise with INV

Progressions: Isometrics → isotonic → single-heel rise

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Longus

ID: ANK-MUS-05

Muscle: Peroneus Longus

Origin: Proximal fibula

Insertion: Medial cuneiform/1st MT plantar

Innervation: Superficial peroneal L5-S1

Blood Supply: Fibular perforators

Function: Eversion + 1st ray PF

EMG: High cutting/uneven ground

Clinical Importance: Tendinopathy; 1st ray stability

Exercises: Band eversion, PL-biased heel raises

Progressions: Isometrics → functional uneven → COD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Brevis

ID: ANK-MUS-06

Muscle: Peroneus Brevis

Origin: Distal fibula

Insertion: 5th MT base

Innervation: Superficial peroneal L5-S1

Blood Supply: Fibular perforators

Function: Eversion; lateral dynamic stability

EMG: High eversion tasks

Clinical Importance: Most torn peroneal tendon

Exercises: Resisted eversion

Progressions: Add unstable surfaces → cutting

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: FHL

ID: ANK-MUS-07

Muscle: FHL

Origin: Posterior fibula

Insertion: Hallux distal phalanx

Innervation: Tibial S2-S3

Blood Supply: Fibular/PTA

Function: Hallux flexion; push-off

EMG: High dance push-off

Clinical Importance: Posterior impingement continuum

Exercises: Toe curls, calf raise with toe flexion

Progressions: Load manage → progressive pointe/push-off

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: FDL

ID: ANK-MUS-08

Muscle: FDL

Origin: Posterior tibia

Insertion: Distal phalanges 2-5

Innervation: Tibial S2-S3

Blood Supply: PTA

Function: Toe flexion; PF/INV assist

EMG: Stance grip

Clinical Importance: Tarsal tunnel contents

Exercises: Towel curls

Progressions: Combined intrinsic training

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: EHL

ID: ANK-MUS-09

Muscle: EHL

Origin: Fibula/IO

Insertion: Hallux distal phalanx dorsal

Innervation: Deep peroneal L5

Blood Supply: Anterior tibial

Function: Hallux extension; DF assist

EMG: Swing

Clinical Importance: L5 myotome

Exercises: Resisted hallux extension

Progressions: Gait clearance drills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: EDL

ID: ANK-MUS-10

Muscle: EDL

Origin: Tib/fib/IO

Insertion: Toes 2-5 extensor expansions

Innervation: Deep peroneal L4-L5

Blood Supply: Anterior tibial

Function: Toe extension; DF assist

EMG: Swing

Clinical Importance: Anterior compartment

Exercises: Heel walks, toe extension

Progressions: DF endurance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibularis Tertius

ID: ANK-MUS-11

Muscle: Fibularis Tertius

Origin: Distal fibula/EDL continuum (variable)

Insertion: 5th MT base dorsal

Innervation: Deep peroneal L5-S1

Blood Supply: Anterior tibial

Function: DF + eversion

EMG: Variable EMG

Clinical Importance: Anatomic variant often present

Exercises: DF-eversion band

Progressions: Uneven ground progressions

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

6. Biomechanics

RECORD: Dorsiflexion

ID: ANK-BIO-01

Topic: Dorsiflexion

Plane: Sagittal

Primary Movers: TA, EHL, EDL, fibularis tertius

Stabilizers: Mortise congruence; ATFL slackens relative

Arthrokinematics: Talus posterior glide (open chain)

Osteokinematics: ~20 deg typical

Functional Role: Ankle rocker midstance; squat depth

Clinical Relevance: Limited DF → early heel rise, Achilles stress, ACL risk cascade proximal

Exercises: Knee-to-wall DF, soleus stretch, step DF

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Plantarflexion

ID: ANK-BIO-02

Topic: Plantarflexion

Plane: Sagittal

Primary Movers: Gastroc, soleus, TP, PL, FHL, FDL

Stabilizers: Anterior capsule/ATFL check extremes

Arthrokinematics: Talus anterior glide (open chain)

Osteokinematics: ~50 deg typical

Functional Role: Propulsion toe-off

Clinical Relevance: Achilles overload; PF weakness impairs push-off

Exercises: Heel raises progressive

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Inversion

ID: ANK-BIO-03

Topic: Inversion

Plane: Frontal / triplanar STJ

Primary Movers: TP, TA, FHL, FDL

Stabilizers: ATFL/CFL lateral check

Arthrokinematics: STJ calcaneal INV open chain

Osteokinematics: ~20 deg STJ contribution approx

Functional Role: Supination component

Clinical Relevance: Lateral sprain mechanism when excessive + PF

Exercises: Controlled INV ROM; avoid early end-range post-sprain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Eversion

ID: ANK-BIO-04

Topic: Eversion

Plane: Frontal / triplanar STJ

Primary Movers: PL, PB, fibularis tertius

Stabilizers: Deltoid medial check

Arthrokinematics: STJ calcaneal EV

Osteokinematics: ~10 deg approx

Functional Role: Pronation component; shock absorption

Clinical Relevance: Deltoid/medial injury if forced; fibularis strengthen for CAI

Exercises: Band eversion

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pronation

ID: ANK-BIO-05

Topic: Pronation

Plane: Triplanar (EV + ABD + DF components)

Primary Movers: Controlled eccentrically by TP/soleus complex

Stabilizers: Spring/TP static-dynamic

Arthrokinematics: STJ + midtarsal unlocking

Osteokinematics: Closed-chain tibial IR coupling

Functional Role: Shock absorb early stance

Clinical Relevance: Excess → PTTD/MTSS cascade; insufficient → poor shock share

Exercises: Short-foot, TP eccentrics, cadence

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Supination

ID: ANK-BIO-06

Topic: Supination

Plane: Triplanar (INV + ADD + PF)

Primary Movers: TP concentric; fibularis eccentrics

Stabilizers: Lateral ligaments

Arthrokinematics: STJ + midtarsal locking late stance

Osteokinematics: Rigid lever for push-off

Functional Role: Propulsive lever

Clinical Relevance: CAI with recurrent INV-supination injury

Exercises: Fibularis strength + NMC

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Closed Chain Mechanics

ID: ANK-BIO-07

Topic: Closed Chain Mechanics

Plane: Multiplanar

Primary Movers: Entire kinetic chain

Stabilizers: Mortise + STJ + midfoot

Arthrokinematics: Tibial rotation drives STJ; opposite open-chain descriptions

Osteokinematics: Foot fixed — body moves over foot

Functional Role: Gait, squat, land

Clinical Relevance: Most sport injuries closed-chain INV

Exercises: Single-leg strength and land training

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Open Chain Mechanics

ID: ANK-BIO-08

Topic: Open Chain Mechanics

Plane: Multiplanar

Primary Movers: Ankle movers

Stabilizers: Ligaments

Arthrokinematics: Classic convex-talus rules

Osteokinematics: Foot free — non-WB ROM testing

Functional Role: Assessment & early rehab

Clinical Relevance: Do not assume open-chain gains transfer without closed-chain training

Exercises: Combine OKC + CKC

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Walking

ID: ANK-BIO-09

Topic: Walking

Plane: Sagittal + STJ triplanar

Primary Movers: TA IC; soleus mid; gastroc/soleus toe-off

Stabilizers: Mortise/STJ

Arthrokinematics: Three rockers

Osteokinematics: DF then PF progression

Functional Role: Community ambulation

Clinical Relevance: Antalgic patterns post-sprain

Exercises: Gait retraining

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Running

ID: ANK-BIO-10

Topic: Running

Plane: Multiplanar

Primary Movers: Soleus/Achilles spring; TA impact

Stabilizers: Bone/tendon capacity

Arthrokinematics: Higher load rates

Osteokinematics: Strike pattern variable

Functional Role: Sport locomotion

Clinical Relevance: Achilles/BSI/CAI risk with spikes

Exercises: Cadence, graded volume

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Jumping

ID: ANK-BIO-11

Topic: Jumping

Plane: Sagittal

Primary Movers: PF power

Stabilizers: Achilles SSC

Arthrokinematics: Rapid PF

Osteokinematics: Triple extension

Functional Role: Jump sports

Clinical Relevance: Achilles tendinopathy if volume excess

Exercises: Plyometric ladder after strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Landing

ID: ANK-BIO-12

Topic: Landing

Plane: Multiplanar

Primary Movers: Eccentric PF/DF control; hip strategy

Stabilizers: CAI risk if INV collapse

Arthrokinematics: Rapid DF loading Achilles/soleus

Osteokinematics: Soft land preferred

Functional Role: ACL/ankle injury prevention crossover

Clinical Relevance: Recurrent sprain on poor land INV

Exercises: Drop-land progressions, fibularis prep

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Cutting

ID: ANK-BIO-13

Topic: Cutting

Plane: Multiplanar

Primary Movers: Fibularis + hip

Stabilizers: ATFL high demand

Arthrokinematics: INV/ER stresses

Osteokinematics: COD

Functional Role: Lateral & high ankle sprain mechanisms

Clinical Relevance: Delayed COD after sprain/syndesmosis

Exercises: Shadow cut → full speed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Balance

ID: ANK-BIO-14

Topic: Balance

Plane: Multiplanar sway

Primary Movers: Intrinsic foot + ankle strategy

Stabilizers: Proprioceptive ligaments

Arthrokinematics: Micro-adjustments STJ/TC

Osteokinematics: SLS

Functional Role: CAI hallmark deficit

Clinical Relevance: Recurrent sprain risk with poor balance

Exercises: Eyes-closed SLS, SEBT/YBT

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Load Transfer

ID: ANK-BIO-15

Topic: Load Transfer

Plane: Axial + shear

Primary Movers: Bone + tendon + ligament share

Stabilizers: Mortise congruence

Arthrokinematics: Tibia → talus → calcaneus/forefoot

Osteokinematics: WB column mechanics

Functional Role: Fracture/OA if malaligned

Clinical Relevance: Restore length/rotation after fracture

Exercises: Progressive WB per injury

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

7. Stability

RECORD: Ankle Stability System

ID: ANK-STAB-01

Static Stabilizers: Bony mortise congruence (wider anterior talus); ATFL, CFL, PTFL; deltoid complex; syndesmosis (AITFL/PITFL/IO); capsule; spring ligament for talar head/medial arch

Dynamic Stabilizers: Peroneus longus/brevis (lateral); TP (medial arch); TA; soleus/gastroc; intrinsic foot muscles

Neuromuscular Control: Feed-forward peroneal activation before landing; delayed peroneal latency common after sprain/CAI

Proprioception: Ligament/capsule mechanoreceptors; deficit after sprain contributes to CAI

Balance: Ankle strategy primary on firm surface; hip strategy when challenged; SEBT/YBT assess

Force Transmission: Axial via plafond; shear restrained by collaterals/syndesmosis; Achilles transmits propulsion

Injury Prevention: Neuromuscular warm-ups, balance programs, external brace/tape in high-risk sport (evidence for sprain prevention), adequate calf capacity, graded COD exposure

Rehabilitation: Restore ROM → strength (especially fibularis/TP/soleus) → balance/NMC → hop/land → cut → RTS criteria

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. International Ankle Consortium CAI criteria themes; sprain prevention literature.

8. Neurovascular

RECORD: Tibial Nerve

ID: ANK-NV-01

Nerve: Tibial nerve (ankle/tarsal tunnel)

Sensory Distribution: Heel via medial calcaneal; plantar foot via medial/lateral plantar

Motor Supply: PF muscles proximal; plantar intrinsics via plantar nerves

Entrapment Sites: Tarsal tunnel behind medial malleolus

Clinical Tests: Tinel at tarsal tunnel; DF-eversion neurodynamics; plantar sensation

Vascular Supply: Travels with posterior tibial artery/vein in tarsal tunnel

Pulses: Posterior tibial pulse — 2 cm posteroinferior to medial malleolus

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deep Peroneal Nerve

ID: ANK-NV-02

Nerve: Deep peroneal (fibular) nerve

Sensory Distribution: 1st web space

Motor Supply: Anterior compartment; EDB/EHB

Entrapment Sites: Anterior tarsal tunnel under extensor retinaculum; anterior CS

Clinical Tests: 1st web sensation; EHL strength; Tinel anterior ankle

Vascular Supply: Courses with anterior tibial → dorsalis pedis

Pulses: Dorsalis pedis — lateral to EHL tendon on dorsum

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Superficial Peroneal Nerve

ID: ANK-NV-03

Nerve: Superficial peroneal nerve

Sensory Distribution: Dorsum foot (most)

Motor Supply: PL/PB

Entrapment Sites: Fascial exit mid-lateral leg; stretched in INV sprain

Clinical Tests: Dorsal sensation; eversion strength; Tinel fascial exit

Vascular Supply: Perforators near course; no major named artery companion throughout

Pulses: Assess DP/PT as limb perfusion context

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sural Nerve

ID: ANK-NV-04

Nerve: Sural nerve

Sensory Distribution: Lateral foot border / posterolateral ankle

Motor Supply: None

Entrapment Sites: Lateral ankle; Achilles surgery; footwear compression

Clinical Tests: Lateral foot sensation; Tinel sural course

Vascular Supply: Near small saphenous vein

Pulses: N/A direct — document DP/PT

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Saphenous Nerve

ID: ANK-NV-05

Nerve: Saphenous nerve

Sensory Distribution: Medial ankle/foot border

Motor Supply: None

Entrapment Sites: Medial ankle surgery; along great saphenous vein

Clinical Tests: Medial sensation; infrapatellar branch history at knee

Vascular Supply: Along great saphenous vein

Pulses: Near but distinct from PT pulse

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

9. Functional Assessment

RECORD: Ankle Functional Assessment

ID: ANK-FUNC-01

ROM: DF (knee flexed/extended), PF, INV, EV; compare sides; knee-to-wall DF (cm)

Strength: Heel-rise endurance (soleus/gastroc), DF MMT, eversion/inversion handheld dynamometry preferred

Gait: Observe IC foot slap, midstance pronation, heel rise symmetry, antalgic patterns

Balance: SLS eyes open/closed; foam; SEBT/YBT

Functional Tests: Heel walking, toe walking, lunges, step-downs

Hop Tests: Single hop, triple hop, side hop, figure-8 — LSI ≥90% often targeted for RTS

Heel Raise: Bilateral then unilateral; note inversion (TP) and height symmetry

Landing: Drop land quality — avoid INV collapse

Return-to-Sport Criteria: Full ROM, strength LSI, hop battery, sport-specific COD without apprehension, psych readiness; time alone insufficient

Outcome Measures: FAAM, FAOS, CAIT (CAI), VISA-A (Achilles), NPRS

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. International Ankle Consortium; VISA-A.

10. Special Tests

RECORD: Anterior Drawer (Ankle)

ID: ANK-TEST-01

Test: Anterior Drawer (Ankle)

Purpose: ATFL integrity / anterior talar translation

Positive Finding: Increased anterior translation ± sulcus sign vs contralateral

Sensitivity: Variable (~30-80% across studies; better delayed than immediate acute)

Specificity: Moderate-high when clearly asymmetric (study-dependent)

Evidence: Best interpreted with history/swelling timing; compare sides

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Talar Tilt

ID: ANK-TEST-02

Test: Talar Tilt

Purpose: CFL (± ATFL) inversion laxity

Positive Finding: Increased inversion talar tilt vs opposite

Sensitivity: Moderate/variable

Specificity: Moderate/variable

Evidence: More useful in chronic CAI; painful acutely

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: External Rotation Stress Test (Kleiger)

ID: ANK-TEST-03

Test: External Rotation Stress Test (Kleiger)

Purpose: Syndesmosis and/or deltoid

Positive Finding: Pain over syndesmosis ± medial clear space feeling

Sensitivity: Moderate

Specificity: Relatively higher specificity reported in some series

Evidence: Part of syndesmosis cluster with squeeze

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Squeeze Test

ID: ANK-TEST-04

Test: Squeeze Test

Purpose: Syndesmosis injury

Positive Finding: Pain at syndesmosis when compressing mid-tibia-fibula

Sensitivity: Lower sensitivity

Specificity: Higher specificity in several studies

Evidence: Positive strongly suggestive; negative does not rule out

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Cotton Test

ID: ANK-TEST-05

Test: Cotton Test

Purpose: Syndesmotic instability (often intraoperative)

Positive Finding: Increased lateral translation of talus in mortise

Sensitivity: Primarily surgical assessment

Specificity: Intraoperative utility

Evidence: Clinical outpatient use limited vs ER/squeeze

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Thompson Test

ID: ANK-TEST-06

Test: Thompson Test

Purpose: Achilles continuity

Positive Finding: Absent PF of foot when calf squeezed = rupture

Sensitivity: Very high (~0.96+ often cited)

Specificity: Very high (~0.93+ often cited)

Evidence: Excellent bedside test — still confirm with US/MRI as needed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Silfverskiöld Test

ID: ANK-TEST-07

Test: Silfverskiöld Test

Purpose: Gastroc vs Achilles/soleus tightness

Positive Finding: DF improves with knee flexion = gastroc limited; no change = soleus/Achilles

Sensitivity: Clinical reliability acceptable for decision-making

Specificity: N/A diagnostic Sn/Sp classic

Evidence: Guides stretching bias (straight vs bent knee)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Single Heel Raise

ID: ANK-TEST-08

Test: Single Heel Raise

Purpose: TP function / Achilles capacity

Positive Finding: Inability to invert/rise or early fatigue — TP dysfunction stages

Sensitivity: Useful clinically for PTTD staging

Specificity: Not a classic Sn/Sp metric

Evidence: Core PTTD examination element

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Star Excursion Balance Test (SEBT)

ID: ANK-TEST-09

Test: Star Excursion Balance Test (SEBT)

Purpose: Dynamic balance / reach asymmetry

Positive Finding: Asymmetry >~4 cm anterior or composite deficits linked to injury risk in some studies

Sensitivity: Predictive utility moderate — population-dependent

Specificity: Screening tool not diagnostic

Evidence: Widely used in CAI research

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Y-Balance Test

ID: ANK-TEST-10

Test: Y-Balance Test

Purpose: Simplified SEBT — anterior/posteromedial/posterolateral

Positive Finding: Asymmetry and low composite scores associated with injury risk in athletic cohorts

Sensitivity: Moderate predictive validity in published cohorts

Specificity: Screening

Evidence: Practical RTS/balance battery component

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

11. Pathologies

RECORD: Lateral Ankle Sprain

ID: ANK-PATH-01

Condition: Lateral Ankle Sprain

Epidemiology: Most common sports ankle injury; ATFL ± CFL

Mechanism: PF-inversion

Symptoms: Lateral pain, swelling, bruising, difficulty WB

Differential Diagnosis: Fracture, syndesmosis, peroneal tear, OLT

Clinical Tests: Anterior drawer, talar tilt, Ottawa rules

Imaging: XR if Ottawa positive; MRI if refractory/CAI

Conservative Treatment: Early protected motion, brace, progressive NMC — avoid prolonged immobilization

Surgical Treatment: Rare acute; reconstruction for CAI failures

Rehabilitation: Functional rehab phases

Return to Sport: Days–weeks grade-dependent with criteria

Prognosis: Good but CAI develops in a substantial minority without adequate rehab

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: High Ankle Sprain (Syndesmosis)

ID: ANK-PATH-02

Condition: High Ankle Sprain (Syndesmosis)

Epidemiology: Less common than lateral; more time-loss

Mechanism: ER-DF; contact

Symptoms: Pain above joint line, worse with ER/DF, out of proportion to swelling sometimes

Differential Diagnosis: Lateral sprain, Maisonneuve, deltoid

Clinical Tests: Squeeze, ER stress, Cotton

Imaging: MRI; WB XR; full-length tib-fib

Conservative Treatment: Protect longer; boot period common

Surgical Treatment: Screw/suture-button if unstable diastasis

Rehabilitation: Delayed cutting vs lateral sprain

Return to Sport: Often 6-8+ weeks minimum

Prognosis: Good if recognized; missed → OA/instability

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deltoid Ligament Sprain

ID: ANK-PATH-03

Condition: Deltoid Ligament Sprain

Epidemiology: Less isolated; often with fibular fracture/syndesmosis

Mechanism: Eversion/ER

Symptoms: Medial pain, medial clear space issues

Differential Diagnosis: Medial malleolus fracture, TP injury, spring ligament

Clinical Tests: Eversion stress, Kleiger

Imaging: Mortise XR, MRI

Conservative Treatment: Protect eversion; address combined injuries

Surgical Treatment: Repair/reconstruct in selected unstable/fracture cases

Rehabilitation: Medial stability + TP

Return to Sport: Tied to combined injury

Prognosis: Depends on associated mortise stability

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Chronic Ankle Instability (CAI)

ID: ANK-PATH-04

Condition: Chronic Ankle Instability (CAI)

Epidemiology: Common sequela after recurrent sprain

Mechanism: Repeated INV injuries + sensorimotor deficit

Symptoms: Giving-way, recurrent sprain, residual pain, impaired balance

Differential Diagnosis: OLT, peroneal tear, subtalar instability, impingement

Clinical Tests: CAIT questionnaire, drawer/tilt, SEBT

Imaging: MRI ligaments/OLT

Conservative Treatment: NMC, strengthening, brace for sport

Surgical Treatment: Anatomic reconstruction (e.g. Broström) if failed care

Rehabilitation: Long-term NMC maintenance

Return to Sport: Criteria + psych

Prognosis: Improved with multimodal care; some need surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Achilles Tendinopathy

ID: ANK-PATH-05

Condition: Achilles Tendinopathy

Epidemiology: Common in runners/masters athletes

Mechanism: Load exceeding capacity — midportion vs insertional

Symptoms: Morning stiffness, load-related tendon pain

Differential Diagnosis: Rupture partial, bursitis, radiculopathy, DVT

Clinical Tests: Palpation, painful arc, Royal London, VISA-A

Imaging: US/MRI if unclear

Conservative Treatment: Education + progressive loading (strong evidence midportion)

Surgical Treatment: Rare for pure tendinopathy; debridement selected insertional

Rehabilitation: Isometrics → HSR → energy storage

Return to Sport: When energy-storage tolerated

Prognosis: Good but often months

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Achilles Rupture

ID: ANK-PATH-06

Condition: Achilles Rupture

Epidemiology: Middle-aged recreational athletes classic peak

Mechanism: Sudden push-off / eccentric DF

Symptoms: Pop, weak PF, palpable gap, difficulty toe-off

Differential Diagnosis: Tennis leg, plantaris, ankle sprain

Clinical Tests: Thompson (high Sn/Sp), Matles, gap

Imaging: US/MRI confirm

Conservative Treatment: Functional rehab protocols with early protected motion accepted option in many centers

Surgical Treatment: Surgical repair selected (athletes/shared decision) — re-rupture/infection tradeoffs

Rehabilitation: Progressive PF loading per protocol 0-6 months

Return to Sport: Often 6-9+ months

Prognosis: Generally good; re-rupture risk exists either path

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Posterior Tibial Tendinopathy (PTTD)

ID: ANK-PATH-07

Condition: Posterior Tibial Tendinopathy (PTTD)

Epidemiology: Middle-aged women common; associated obesity

Mechanism: Degenerative overload of TP/spring

Symptoms: Medial ankle pain, progressive flatfoot, difficulty heel rise

Differential Diagnosis: Spring tear, deltoid, midfoot OA, tarsal coalition youth

Clinical Tests: Single heel rise, too-many-toes, too many toes + valgus

Imaging: MRI/US; WB XR alignment

Conservative Treatment: Stage-based orthoses, loading, PT early

Surgical Treatment: Tendon transfer/osteotomy/arthrodesis advanced

Rehabilitation: Stage I-II active care; III-IV structural

Return to Sport: Limited high-impact if advanced flatfoot

Prognosis: Stage-dependent

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneal Tendinopathy

ID: ANK-PATH-08

Condition: Peroneal Tendinopathy

Epidemiology: Runners, dancers, post-sprain

Mechanism: Repetitive eversion / instability / cavovarus

Symptoms: Lateral retromalleolar pain, swelling

Differential Diagnosis: Lateral sprain, sural nerve, 5th MT fracture

Clinical Tests: Resisted eversion, subluxation observation, US

Imaging: US dynamic; MRI tears

Conservative Treatment: Load + stability; address retinaculum

Surgical Treatment: Repair/groove deepening for dislocation/tears

Rehabilitation: Progressive eversion + NMC

Return to Sport: When COD tolerated

Prognosis: Good if instability fixed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Anterior Ankle Impingement

ID: ANK-PATH-09

Condition: Anterior Ankle Impingement

Epidemiology: Soccer athletes classic (athlete's ankle)

Mechanism: Repetitive DF; osteophytes

Symptoms: Anterior pain with DF, kicking

Differential Diagnosis: OLT, soft-tissue impingement, ankle OA

Clinical Tests: DF impingement provocation

Imaging: XR osteophytes; MRI soft tissue

Conservative Treatment: Activity mod, DF mobility carefully, injection selected

Surgical Treatment: Arthroscopic debridement

Rehabilitation: Restore DF without forcing bony block

Return to Sport: When DF sport demands met

Prognosis: Good after arthroscopy if OA limited

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Posterior Ankle Impingement

ID: ANK-PATH-10

Condition: Posterior Ankle Impingement

Epidemiology: Ballet/pointe, kicking sports

Mechanism: Os trigonum / FHL / posterior process

Symptoms: Posterior pain with PF

Differential Diagnosis: Achilles insertional, Haglund, FHL tendinopathy

Clinical Tests: PF impingement, posteromedial/lateral palpation

Imaging: XR os trigonum; MRI

Conservative Treatment: Load modify, FHL mobility, injection

Surgical Treatment: Excise os trigonum selected

Rehabilitation: Graded PF loading

Return to Sport: When PF sport demands met

Prognosis: Good with activity adjust/surgery selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Osteochondral Lesion of Talus (OLT)

ID: ANK-PATH-11

Condition: Osteochondral Lesion of Talus (OLT)

Epidemiology: Post-sprain or idiopathic; medial or lateral talar dome

Mechanism: Trauma/ischemia theories

Symptoms: Deep ankle pain, catching, swelling after activity

Differential Diagnosis: Impingement, loose body, OA

Clinical Tests: Pain with load; limited specific test

Imaging: MRI staging; CT for cystic

Conservative Treatment: Unload, protected WB selected lesions

Surgical Treatment: Microfracture/OATS/allograft per size/location

Rehabilitation: Protected WB → progressive impact per surgery

Return to Sport: Delayed months

Prognosis: Variable by lesion size/stability

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Ankle Osteoarthritis

ID: ANK-PATH-12

Condition: Ankle Osteoarthritis

Epidemiology: Post-traumatic common etiology

Mechanism: Prior fracture/instability/OLT

Symptoms: Activity pain, stiffness, swelling, reduced DF

Differential Diagnosis: Impingement, inflammatory arthritis

Clinical Tests: ROM, crepitus, alignment

Imaging: WB XR

Conservative Treatment: Exercise, weight loss, brace, injection selective

Surgical Treatment: Arthrodesis/arthroplasty selected end-stage

Rehabilitation: Maintain function, strength, activity pacing

Return to Sport: Modified sport often

Prognosis: Chronic progressive

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sinus Tarsi Syndrome

ID: ANK-PATH-13

Condition: Sinus Tarsi Syndrome

Epidemiology: Post-sprain lateral hindfoot pain syndrome

Mechanism: STJ instability / synovitis in sinus tarsi

Symptoms: Lateral pain worse on uneven ground

Differential Diagnosis: CFL, CECS, peroneal, STJ OA

Clinical Tests: Sinus tarsi tenderness, STJ motion pain

Imaging: MRI sinus tarsi; anesthetic injection diagnostic

Conservative Treatment: Proprioception, STJ control, injection

Surgical Treatment: Synovectomy rare

Rehabilitation: Balance + fibularis/TP

Return to Sport: When uneven ground tolerated

Prognosis: Variable

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Ankle/Hindfoot Stress Fracture

ID: ANK-PATH-14

Condition: Ankle/Hindfoot Stress Fracture

Epidemiology: Navicular high-risk; medial malleolus; calcaneus in runners/military

Mechanism: Repetitive overload / energy deficit

Symptoms: Focal bony pain, night pain, hop pain

Differential Diagnosis: Tendinopathy, sprain, MTSS

Clinical Tests: Point tenderness, hop

Imaging: MRI preferred early

Conservative Treatment: Unload risk-stratified; nutrition/RED-S screen

Surgical Treatment: Navicular may need fixation if delayed/high-risk

Rehabilitation: Protected → graded run

Return to Sport: Delayed; site-dependent

Prognosis: Guarded for navicular tension-side

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Ankle Region Fractures (Malleolar, Talar, Calcaneal)

ID: ANK-PATH-15

Condition: Ankle Region Fractures (Malleolar, Talar, Calcaneal)

Epidemiology: Common traumatic injuries; classification guides care (Weber/Lauge-Hansen; Hawkins; Sanders)

Mechanism: Twisting, fall from height (calcaneus), forced DF/PF (talus)

Symptoms: Pain, swelling, inability to WB, deformity

Differential Diagnosis: Severe sprain, dislocation, syndesmosis

Clinical Tests: Ottawa rules; neurovascular exam mandatory

Imaging: XR; CT articular mapping

Conservative Treatment: Stable nondisplaced may be nonop in boot/cast per ortho

Surgical Treatment: ORIF common for unstable malleolar; talar neck urgent reduction; calcaneus individualized

Rehabilitation: Surgeon WB/ROM protocols; progressive strength/gait; watch for AVN after talar neck

Return to Sport: Months; articular injuries longer

Prognosis: Depends on reduction quality; OA risk residual

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

12. Rehabilitation

RECORD: Lateral Ankle Sprain Rehab

ID: ANK-REHAB-01

Condition: Lateral ankle sprain

Acute Phase: Protect, optimal loading, ice/compress/elevate as needed; rule out fracture (Ottawa)

Protection: Functional brace preferred over prolonged cast for most grades I-II

Mobility: Early DF/PF within pain; progress INV carefully

Strength: Isometrics → isotonic fibularis/TP/calf

Balance: SLS progressions early as pain allows

Plyometrics: After strength/balance base — hop in place → multiplanar

Running Progression: Walk-run when hop comfortable

Cutting: Last — after linear run and land quality

Sport-Specific Drills: Graded COD and sport skills

Return-to-Play Criteria: Full ROM, strength, hop LSI, sport drills without giving-way; brace optional for high-risk

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Achilles Tendinopathy Rehab

ID: ANK-REHAB-02

Condition: Achilles tendinopathy

Acute Phase: Reduce provocative sprint/hill volume; maintain load capacity

Protection: Avoid complete rest; relative rest from aggravators

Mobility: Gentle DF; careful with insertional (limit aggressive stretch into compression)

Strength: Isometrics for analgesia → heavy slow resistance → energy storage

Balance: SLS as accessory

Plyometrics: Late — after HSR tolerance

Running Progression: Pain-monitoring return

Cutting: After linear run tolerance

Sport-Specific Drills: Graded sprint/jump volumes

Return-to-Play Criteria: VISA-A improvement; sport loads tolerated repeatedly

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Cook & Purdam; Alfredson/HSR literature.

RECORD: Achilles Rupture Rehab

ID: ANK-REHAB-03

Condition: Achilles rupture (functional or post-op)

Acute Phase: Ortho protocol — protected PF position early often

Protection: Boot with wedges weaning per protocol

Mobility: Gradual DF toward neutral then beyond per timeline

Strength: Seated PF → bilateral standing → unilateral late

Balance: When WB allows

Plyometrics: Late (often 4-6+ months)

Running Progression: Often ~4-6 months if criteria met

Cutting: After run and hop capacity

Sport-Specific Drills: Graded 6-9+ months

Return-to-Play Criteria: Heel-rise index, hop, psych, surgeon clearance — often ≥6-9 months

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: CAI Rehab

ID: ANK-REHAB-04

Condition: Chronic ankle instability

Acute Phase: Settle irritability if recent sprain

Protection: Brace for high-risk sport during rehab

Mobility: Restore DF if limited

Strength: Fibularis, TP, proximal hip

Balance: Progressive NMC cornerstone — SEBT/YBT

Plyometrics: Multiplanar landings

Running Progression: As control allows

Cutting: Emphasize technique + fatigue resistance

Sport-Specific Drills: Reactive COD

Return-to-Play Criteria: CAIT improvement, hop/YBT, no giving-way in sport drills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. International Ankle Consortium.

RECORD: Syndesmosis Rehab

ID: ANK-REHAB-05

Condition: High ankle sprain

Acute Phase: Longer protection than lateral; boot common

Protection: Avoid early forced ER/DF

Mobility: Gradual DF after protection window

Strength: Calf, fibularis, proximal

Balance: Delayed vs simple sprain

Plyometrics: Delayed

Running Progression: Later — often after 4-6 weeks minimum depending on grade

Cutting: Last — ER stress risk

Sport-Specific Drills: Graded after linear power

Return-to-Play Criteria: Pain-free ER/DF, hop, sport drills; imaging/surgeon if fixed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: PTTD Rehab

ID: ANK-REHAB-06

Condition: Posterior tibial tendinopathy / early AAFD

Acute Phase: Unload medial tendon (orthosis/boot as stage dictates)

Protection: UCBL/AFO/orthoses stage-based

Mobility: Gastroc-soleus flexibility if limited DF contributes to pronation

Strength: TP progressive, intrinsics, proximal hip

Balance: Arch-control SLS

Plyometrics: Only early stages if appropriate

Running Progression: Often limited in advanced flatfoot

Cutting: Caution in flexible collapse

Sport-Specific Drills: Individualize to stage

Return-to-Play Criteria: Single heel rise capacity, pain, orthosis plan for sport

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

13. Evidence

RECORD: Key Evidence — Ankle Module

ID: ANK-EVID-01

Topic: Foundational ankle references

Anatomy_Biomechanics: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

Ottawa_Rules: Validated decision rules to reduce unnecessary ankle/foot radiographs after acute injury.

CAI: International Ankle Consortium selection criteria and CAIT themes for chronic ankle instability research/clinical framing.

Achilles: Thompson test excellent accuracy; progressive loading for tendinopathy; shared decision functional vs surgical rupture care.

Prevention: Neuromuscular programs and bracing can reduce sprain rates in high-risk sports (evidence supportive).

AI_Note: Sn/Sp and timelines are approximate and study-dependent — verify with current protocols and individual assessment.

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.